# Dataset Notes

Capstone Client Handoff: Final Validated Dataset

## File Included

final_validated_dataset.csv

## Purpose

This file is the final cleaned and merged dataset used for the capstone modeling deliverables. It supports both Task A current GSTD classification and Task B 7-day GSTD 5 alert modeling.

## Dataset Origin

The dataset was created from DSSAT output files provided for the project. The main source files included:

- PlantGro.OUT
- SoilNi.OUT
- SoilWat.OUT
- Weather.OUT

These files were parsed into tabular data using the team's DSSAT output parser notebook:

- DSSATout2Dataframe.ipynb

The step-by-step workflow for parsing, cleaning, merging, validating, and saving the dataset is documented in the appendix notebook:

- Dataset_Creation_and_Validation_Workflow.ipynb

During development, the validated merged dataset was saved as capstone_secondary_dssat_merged_validated.csv. For the client handoff, the same final validated modeling dataset was standardized under the name final_validated_dataset.csv so the final notebooks and handoff documents use one consistent file name.

## Dataset Creation Workflow Summary

At a high level, the dataset creation workflow followed these steps:

- Parse the raw DSSAT .OUT files into dataframes.
- Standardize identifiers and date-level keys across parsed outputs.
- Merge the plant growth, soil nitrogen, soil water, and weather tables using run, treatment, year, and day-of-year level keys.
- Check for duplicate run-day records, expected run and treatment counts, year coverage, and missingness.
- Clean redundant metadata columns and save the validated merged modeling dataset.

## Dataset Scope

The dataset contains DSSAT-derived simulation records after parsing, cleaning, merging, and validation by the project team. The final modeling notebooks assume this file name exactly: final_validated_dataset.csv.

## How the Dataset Is Used

- Task A uses the dataset to classify current growth stage (GSTD).
- Task B uses the dataset to construct a 7-day transition alert target for entry into GSTD 5.
- RUNNO is used for grouped validation to avoid leakage across repeated observations from the same simulation run.
- Year-based splits are used for forward-time validation.

- Treatment identifiers are used for treatment-holdout robustness checks.

## Important Notes

- The dataset is DSSAT/simulation-derived and should be validated against real operational or field data before production deployment.
- The final notebooks expect the dataset to be placed either in the same folder as the notebook or in ../02_Data/.
- Client-provided variable descriptions are not repeated here. This document only explains how the dataset was produced and how it is used in the modeling handoff.